# MIDTERM TEST - 2026

*Course:* **CS 3332 – Software Engineering**
*Date:* *17th April, 2026* *Duration: 90 minutes*
*Total of pages:* *Total marks: 100*

*Student name:* .....

*Student ID:* .....

*Note:* **- Open book (Students can use any written publication or copy);**
**- Internetworking devices (mobiles, tablet computers, watches, and others) are not permitted.**

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**Q1. Multiple choice questions (20x2 points) - Cycle the correct answer.**

**1.1. The Waterfall model is characterized by:**
- Sequential phases, little iteration
- Flexible, continuous changes
- No documentation required
- Only used for small software

**1.2. A good software process requires:**
- Clear documentation
- Customer involvement
- Testing
- All of the above

**1.3. In Agile, a "Sprint" is:**
- A requirements document
- A testing tool
- A short development cycle
- A type of design

**1.4. In Waterfall, if requirements change then:**
- Easy to adapt
- Difficult, costly
- No impact
- No need to care

**1.5. Requirement is:**
- Requirement of systems
- Customer's demand for the software
- User interface design
- Testing document

**1.6. Requirement specification is:**
- User interface design
- A technical requirement
- A document describing software requirements in detail
- Test plan

**1.7. Software architecture design describes:**
- Overall structure and main components of the system
- Architecture of individual programs
- System testing
- User interface design



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**1.8. Structural design helps:**

- Write code faster
- Ensure the system is easy to maintain and extend
- No testing needed
- No customer needed

**1.9. An important characteristic of software engineering is:**

- Just writing code
- No testing needed
- Focus on process and quality
- No customer needed

**1.10. Why is software often harder to maintain than hardware?**

- Software is easy to change and requirements often fluctuate
- Software cannot be fixed
- Software has no documentation
- Software is unrelated to customers

**1.11. The Spiral model combines:**

- No testing needed
- Fast coding
- Iterative development and risk management
- No customer needed

**1.12. Scrum defines main roles as:**

- Tester, Designer, Manager
- Customer, Developer, Analyst
- Programmer, Tester, Architect
- Product Owner, Scrum Master, Development Team

**1.13. Product backlog is:**

- Detailed code
- User interface design
- A list of all product requirements and features
- Testing document

**1.14. Use case helps:**

- Write detailed code
- Design database
- Test the system
- Describe interactions between users and the system

**1.15. State diagram describes:**

- The system's responsese
- Database design
- Software testing
- System states and transitions between them

**1.16. During requirement elicitation, if you only rely on interviewing a small group of users, the risk is:**

- No impact
- No need for testing
- No need for customers
- Requirements are incomplete and lack representativeness

**1.17. In an activity diagram, if there are many complex loops, the risk is:**

- Reduced performance
- Difficult testing and maintenance
- No customer needed
- No impact

**1.18. Compare Waterfall and Iterative models:**

- Waterfall is sequential, Iterative repeats and improves gradually
- Waterfall is flexible, Iterative is fixed
- Both require no documentation
- No difference

**1.19. Compare Class diagram and Component diagram:**

- Class diagram shows classes and relationships, Component diagram shows software component structure
- Class diagram shows UI, Component diagram shows data
- Both are the same
- Not needed in modelling

**1.20. Compare Layered and Client-Server architectures:**

- Layered has clear separation of layers, Client-Server focuses on client–server relationship
- Layered only for small systems, Client-Server for large systems
- Both are the same
- Not needed in design

**Q2. (20 points):** Using the technique suggested here, where natural language descriptions are presented in a standard format, write plausible user requirements for the following function: An unattended petrol (gas) pump system that includes a credit card reader. The customer swipes the card through the reader, then specifies the amount of fuel required. The fuel is delivered and the customer's account debited.

**Q3. (20 points):** Draw state diagrams of the control software for an automatic washing machine that has different programs for different types of clothes.

**Q4.** The dean of the Faculty of Engineering asks a company to develop a student management system for his faculty. Do the following:

**Q4.1. (10 points):** List 03 functional and 02 non-functional requirements.

**Q4.2. (10 points):** Draw 05 use case diagrams of the system. Among them, write 02 textual descriptions of use cases.



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